

MUHAMMAD ADEEL

AI/ML Engineer | Generative AI | LLMs | RAG | AI Agents

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SUMMARY

AI/ML engineer building systems across machine learning, deep learning, and generative AI — from transfer-learning computer vision models to retrieval-augmented LLM applications and tool-calling AI agents. Experienced across the ML lifecycle, including data preprocessing, model training and evaluation, and API-based deployment with FastAPI, Flask, and Docker. Works fluently in Python, PyTorch/TensorFlow, and LangChain/LangGraph.

TECHNICAL SKILLS

Programming & Data: Python, SQL, MySQL, Pandas, NumPy

Machine Learning: Scikit-learn, XGBoost, LightGBM, CatBoost, Regression, Classification, Ensemble Learning, Feature Engineering, Cross-Validation

Deep Learning & NLP: PyTorch, TensorFlow, Transformers, NLP, CNN, RNN, GNN, Transfer Learning

Generative AI & LLMs: LLMs, RAG, Embeddings, Vector Search, Semantic Search, Prompt Engineering, LangChain, LangGraph, AI Agents, Tool Calling

Computer Vision: EfficientNetV2, DenseNet121, YOLO, BLIP, OCR, Image Classification, Object Detection, Image Captioning

Backend & Deployment: FastAPI, Flask, REST APIs, Docker, Git, GitHub

PROFESSIONAL EXPERIENCE

Data Analyst — Positive Solutions, Wah Cantt

Apr 2026 – Jul 2026

- Analyzed business datasets using Excel, MySQL, and Python to support recurring cross-department reporting
- Built and maintained dashboards to track key performance metrics and surface trends for stakeholders
- Wrote and optimized SQL queries (joins, aggregations, subqueries) to extract and clean data, improving reporting accuracy
- Automated repetitive data-processing tasks with Python, reducing manual effort and speeding up decision-making

Artificial Intelligence Trainee — NAVTTC, Islamabad

Oct 2025 – Mar 2026

- Trained in core AI/ML concepts, assisting in developing and testing models using Python, NumPy, and Pandas for data preprocessing and evaluation
- Explored generative AI fundamentals, including LLMs and prompt engineering, alongside hands-on ML/DL model experimentation

PROJECTS

Enterprise AI Knowledge Assistant — RAG / LLM [\[GitHub\]](#)

- Built a retrieval-augmented generation system for querying enterprise documents, covering ingestion, chunking, and embedding generation for a vector search index
- Implemented a retrieval pipeline grounding LLM responses in retrieved context; containerized with Docker and exposed via a FastAPI backend

AI Customer Support Agent — LLM / AI Agents [\[GitHub\]](#)

- Developed an LLM-based conversational agent using LangChain/LangGraph to handle customer queries through multi-step agent workflows
- Implemented tool-calling for database/API interaction, combined with RAG retrieval to ground responses in business knowledge

AI-Powered “Third Eye” for Visually Impaired Persons — Multimodal AI / Computer Vision [\[GitHub\]](#)

- Built a real-time multi-module assistive system combining DenseNet121 (currency classification), YOLO11n (object detection), and BLIP (scene captioning) with OCR and text-to-speech for real-time spoken feedback
- Achieved 90% classification accuracy across the assistive modules on a Pakistan-specific dataset

Spine Disease Detection System — EfficientNetV2 / Final Year Project [\[GitHub\]](#)

- Developed a computer vision system to classify spinal disorders (neural foraminal narrowing, subarticular stenosis, spinal canal stenosis) using the RSNA 2024 Lumbar Spine dataset
- Fine-tuned EfficientNetV2 with transfer learning, data augmentation, and sampling techniques to address class imbalance
- Deployed the model via Flask, achieving 95% classification accuracy

Heart Disease Prediction — Ensemble Machine Learning [\[GitHub\]](#)

- Built and compared classification models (Logistic Regression, Random Forest, SVM, XGBoost, LightGBM, CatBoost) on the Kaggle Playground Series S6E2 dataset, reaching an ensemble AUC of ~0.956 via probability averaging

EDUCATION

Bachelor of Computer Science — University of Haripur

Sep 2021 – Jun 2025

CERTIFICATIONS

- Machine Learning Specialization — Andrew Ng (Coursera) [\[Verify\]](#)
- Deep Learning Specialization — Andrew Ng (Coursera) [\[Verify\]](#)